THE LIMIT OF THE COHOMOLOGY OF THE QUOTIENTS COMPUTES THE COHOMOLOGY OF THE DERIVED COMPLETION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- For a Noetherian ring the derived completion of a complex is the derived limit of the complexes tensored with the quotients by the powers of the ideal. invoked at proof lines 28538-28540. Identifies the object whose cohomology is being computed with the derived limit of the systems appearing in the statement.
- The cohomology of the derived completion of a pseudo-coherent complex is the inverse limit of the cohomology of the quotients. invoked at proof lines 28541-28542. Computes the cohomology of the derived completion as the inverse limit of the cohomology of the powers, which is the left hand side of the statement.
- A short exact sequence of inverse systems gives a short exact sequence relating the derived limit and the inverse limit, with the obstruction term on the left. invoked at proof lines 28543-28547. Produces the displayed short exact sequence whose middle term is the computed cohomology and whose right hand term is the limit in the statement.

Source reference. Original proposition 0922: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27219–27219. Invoked at proof lines 28538-28540 to identify the derived completion; the computation of its cohomology and the exact sequence argument that follow are inside the counted span.

Source reference. Original lemma 0A06: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27266–27266. Used at proof lines 28541-28542 to identify the cohomology of the derived completion with the inverse limit of the cohomology of the quotients; the rest of the argument is inside the counted span.

Source reference. Original lemma 0CQE: sources/stacks-more-algebra/pinned-original-source.tex, source lines 23653–23653. Applied at proof lines 28543-28547 to obtain the displayed exact sequence; the vanishing argument using Mittag-Leffler that finishes the proof is inside the counted span.

Source reference. Original lemma 091D: sources/stacks-more-algebra/pinned-original-source.tex, source lines 23574–23574. Used at proof lines 28548-28549 to see that the left hand term of the exact sequence vanishes; the conclusion drawn from that vanishing is inside the counted span.

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Lemma 0.2 (Kollár-Kovács)**.** *Let I be an ideal of a Noetherian ring A. Let $K \in D(A)$. Set $K_n = K \otimes^{\mathbf{L}}_A A/I^n$. Assume for all $i \in \mathbf{Z}$ we have*

(1) *$H^i(K)$ is a finite A-module, and*
(2) *the system $H^i(K_n)$ satisfies Mittag-Leffler.*

Then $\lim H^i(K)/I^n H^i(K)$ is equal to $\lim H^i(K_n)$ for all $i \in \mathbf{Z}$.

Proof. Recall that $K^\wedge = R\lim K_n$ is the derived completion of K, see Proposition 0922. By Lemma 0A06 we have $H^i(K^\wedge) = \lim H^i(K)/I^n H^i(K)$. By Lemma 0CQE we get short exact sequences

$$0 \to R^1 \lim H^{i-1}(K_n) \to H^i(K^\wedge) \to \lim H^i(K_n) \to 0$$

The Mittag-Leffler condition guarantees that the left terms are zero (Lemma 091D) and we conclude the lemma is true. □